Table 1: PILOT DIAGNOSTIC; NOT MANUSCRIPT RESULTS. Simulation results under retrospective sampling with good overlap ($\psi = 0.6$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 5 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	7.0	12.4	7.0	12.4	1.00	0.03	1.00	0.03	100.0	100.0	100.0	100.0
	PGM	–	×	7.9	7.9	11.8	11.8	0.19	0.19	0.18	0.18	80.0	80.0	80.0	80.0
	DSM	×	×	5.3	5.6	6.6	11.3	0.17	0.18	0.29	0.15	100.0	100.0	100.0	60.0
	SW-PSM	✓	–	−2.1	0.7	−2.1	0.7	0.08	0.21	0.08	0.21	100.0	100.0	100.0	100.0
	WDSM (M=1)	✓	×	−0.2	−0.5	−1.8	7.8	0.28	0.38	0.08	0.04	100.0	100.0	100.0	80.0
	WDSM (M=3)	✓	×	−0.4	−0.6	−0.4	8.0	0.15	0.21	0.12	0.06	100.0	100.0	100.0	80.0
	WDSM (M=5)	✓	×	−0.9	−0.8	−0.5	7.0	0.16	0.18	0.13	0.07	100.0	100.0	100.0	80.0
PATT	PSM	×	–	10.2	17.0	10.2	17.0	1.00	0.37	1.00	0.37	100.0	80.0	100.0	80.0
	PGM	–	×	8.8	8.8	16.3	16.3	2.26	2.26	0.65	0.65	80.0	80.0	20.0	20.0
	DSM	×	×	10.7	9.9	10.2	14.5	2.94	2.86	1.11	1.52	60.0	60.0	60.0	40.0
	SW-PSM	✓	–	4.2	2.0	4.2	2.0	3.16	4.22	3.16	4.22	80.0	100.0	80.0	100.0
	WDSM (M=1)	✓	×	4.6	3.0	4.4	7.9	2.92	1.84	1.21	0.53	100.0	100.0	100.0	80.0
	WDSM (M=3)	✓	×	3.4	2.4	3.8	7.4	3.23	7.50	1.12	1.18	100.0	100.0	100.0	80.0
	WDSM (M=5)	✓	×	3.0	2.7	3.7	7.6	2.55	5.03	2.60	1.02	100.0	100.0	100.0	80.0

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses survey-weighted propensity estimation under retrospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\overline{\widehat{\tau}} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted. **These five-replication pilot diagnostics must not replace manuscript results.**

Table 2: PILOT DIAGNOSTIC; NOT MANUSCRIPT RESULTS. Simulation results under retrospective sampling with poor overlap ($\psi = 2.0$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 5 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	6.3	26.8	6.3	26.8	1.00	0.95	1.00	0.95	100.0	100.0	100.0	100.0
	PGM	–	×	11.9	11.9	20.7	20.7	14.12	14.12	6.89	6.89	60.0	60.0	40.0	40.0
	DSM	×	×	0.9	−0.4	0.0	12.4	5.20	5.51	5.46	4.28	100.0	100.0	100.0	80.0
	SW-PSM	✓	–	−14.2	−16.0	−14.2	−16.0	3.18	2.93	3.18	2.93	80.0	80.0	80.0	80.0
	WDSM (M=1)	✓	×	−4.9	−4.1	−6.0	16.7	5.63	8.96	4.89	4.94	100.0	100.0	100.0	100.0
	WDSM (M=3)	✓	×	−1.3	−0.9	−7.6	19.9	8.10	11.37	7.72	8.15	100.0	100.0	100.0	80.0
	WDSM (M=5)	✓	×	0.1	−0.2	−3.0	24.3	9.31	9.92	7.72	6.48	100.0	100.0	100.0	80.0
PATT	PSM	×	–	−5.0	8.2	−5.0	8.2	1.00	1.37	1.00	1.37	100.0	100.0	100.0	100.0
	PGM	–	×	6.0	6.0	25.5	25.5	3.93	3.93	8.94	8.94	80.0	80.0	0.0	0.0
	DSM	×	×	3.0	1.0	2.5	14.2	4.10	5.79	0.38	0.40	100.0	100.0	100.0	40.0
	SW-PSM	✓	–	−0.6	−1.4	−0.6	−1.4	5.44	7.95	5.44	7.95	100.0	100.0	100.0	100.0
	WDSM (M=1)	✓	×	1.2	0.2	3.9	16.1	7.12	7.40	1.96	1.73	100.0	100.0	100.0	60.0
	WDSM (M=3)	✓	×	1.3	1.4	3.3	16.3	5.83	4.85	4.17	5.38	100.0	100.0	100.0	40.0
	WDSM (M=5)	✓	×	1.3	0.5	3.6	16.3	8.22	6.61	6.29	5.51	100.0	100.0	100.0	60.0

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses survey-weighted propensity estimation under retrospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\overline{\widehat{\tau}} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted. **These five-replication pilot diagnostics must not replace manuscript results.**

Table 3: PILOT DIAGNOSTIC; NOT MANUSCRIPT RESULTS. Simulation results under prospective sampling with good overlap ($\psi = 0.6$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 5 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	7.4	16.0	7.4	16.0	1.00	0.39	1.00	0.39	100.0	80.0	100.0	80.0
	PGM	–	×	8.8	8.8	18.4	18.4	0.44	0.44	0.74	0.74	80.0	80.0	20.0	20.0
	DSM	×	×	8.2	7.9	8.6	13.7	1.11	0.97	0.64	2.73	80.0	80.0	100.0	60.0
	SW-PSM	✓	–	0.8	1.5	0.8	1.5	0.25	1.85	0.25	1.85	100.0	100.0	100.0	100.0
	WDSM (M=1)	×	×	0.9	2.6	3.0	11.1	0.20	0.28	0.93	0.23	100.0	100.0	100.0	80.0
	WDSM (M=3)	×	×	2.6	3.6	4.2	11.6	0.36	0.39	0.42	0.23	100.0	100.0	100.0	80.0
	WDSM (M=5)	×	×	3.2	3.2	4.6	11.5	0.41	0.37	0.36	0.27	100.0	100.0	100.0	80.0
PATT	PSM	×	–	8.1	10.3	8.1	10.3	1.00	1.92	1.00	1.92	100.0	100.0	100.0	100.0
	PGM	–	×	9.7	9.7	18.2	18.2	10.11	10.11	3.70	3.70	60.0	60.0	0.0	0.0
	DSM	×	×	9.0	9.7	10.1	14.3	0.97	2.83	1.00	25.66	60.0	60.0	80.0	60.0
	SW-PSM	✓	–	−0.1	2.5	−0.1	2.5	2.88	3.06	2.88	3.06	100.0	100.0	100.0	100.0
	WDSM (M=1)	×	×	0.3	2.5	3.1	5.5	0.66	1.55	1.51	2.04	100.0	100.0	100.0	100.0
	WDSM (M=3)	×	×	1.2	2.3	2.8	7.6	1.51	2.31	1.89	11.95	100.0	100.0	100.0	100.0
	WDSM (M=5)	×	×	1.8	1.8	2.9	6.7	1.43	1.81	3.55	9.66	100.0	100.0	100.0	100.0

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses unweighted propensity estimation under prospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\overline{\widehat{\tau}} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted. **These five-replication pilot diagnostics must not replace manuscript results.**

Table 4: PILOT DIAGNOSTIC; NOT MANUSCRIPT RESULTS. Simulation results under prospective sampling with poor overlap ($\psi = 2.0$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 5 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	38.4	67.6	38.4	67.6	1.00	0.95	1.00	0.95	100.0	40.0	100.0	40.0
	PGM	–	×	3.2	3.2	23.7	23.7	15.07	15.07	8.60	8.60	100.0	100.0	40.0	40.0
	DSM	×	×	12.6	12.4	13.9	24.6	2.30	2.41	3.72	3.27	80.0	80.0	80.0	40.0
	SW-PSM	✓	–	15.4	11.7	15.4	11.7	1.15	1.30	1.15	1.30	20.0	40.0	20.0	40.0
	WDSM (M=1)	×	×	2.6	3.2	3.0	27.8	5.17	4.50	3.21	3.67	100.0	100.0	100.0	80.0
	WDSM (M=3)	×	×	0.8	2.5	6.1	29.2	7.43	6.80	3.29	2.38	100.0	100.0	80.0	60.0
	WDSM (M=5)	×	×	1.9	2.7	5.2	27.4	9.21	8.06	4.29	2.72	100.0	100.0	80.0	60.0
PATT	PSM	×	–	4.7	19.6	4.7	19.6	1.00	0.43	1.00	0.43	100.0	80.0	100.0	80.0
	PGM	–	×	4.2	4.2	25.5	25.5	6.45	6.45	3.15	3.15	100.0	100.0	0.0	0.0
	DSM	×	×	3.9	2.3	−0.2	18.2	1.33	1.05	16.33	3.03	100.0	100.0	100.0	80.0
	SW-PSM	✓	–	−6.0	−7.0	−6.0	−7.0	0.47	0.32	0.47	0.32	80.0	80.0	80.0	80.0
	WDSM (M=1)	×	×	−3.3	−3.7	−5.2	10.0	1.63	1.04	18.16	2.94	80.0	80.0	100.0	100.0
	WDSM (M=3)	×	×	−3.6	−2.8	−0.5	10.2	1.03	0.76	2.22	1.15	80.0	80.0	100.0	80.0
	WDSM (M=5)	×	×	−4.2	−2.9	−1.5	9.9	0.88	0.64	1.42	0.97	80.0	80.0	100.0	80.0

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses unweighted propensity estimation under prospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\overline{\widehat{\tau}} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted. **These five-replication pilot diagnostics must not replace manuscript results.**